

Week 4 – End-to-End Sales & Profit Analysis

Stakeholder Explanation Report

This project analyzes the provided Financial Sample dataset using data cleaning, exploratory analysis, business aggregation and visualization. The analysis is designed to support stakeholder decisions about sales performance, profitability, products, countries, customer segments and time trends.

1. Dataset Overview

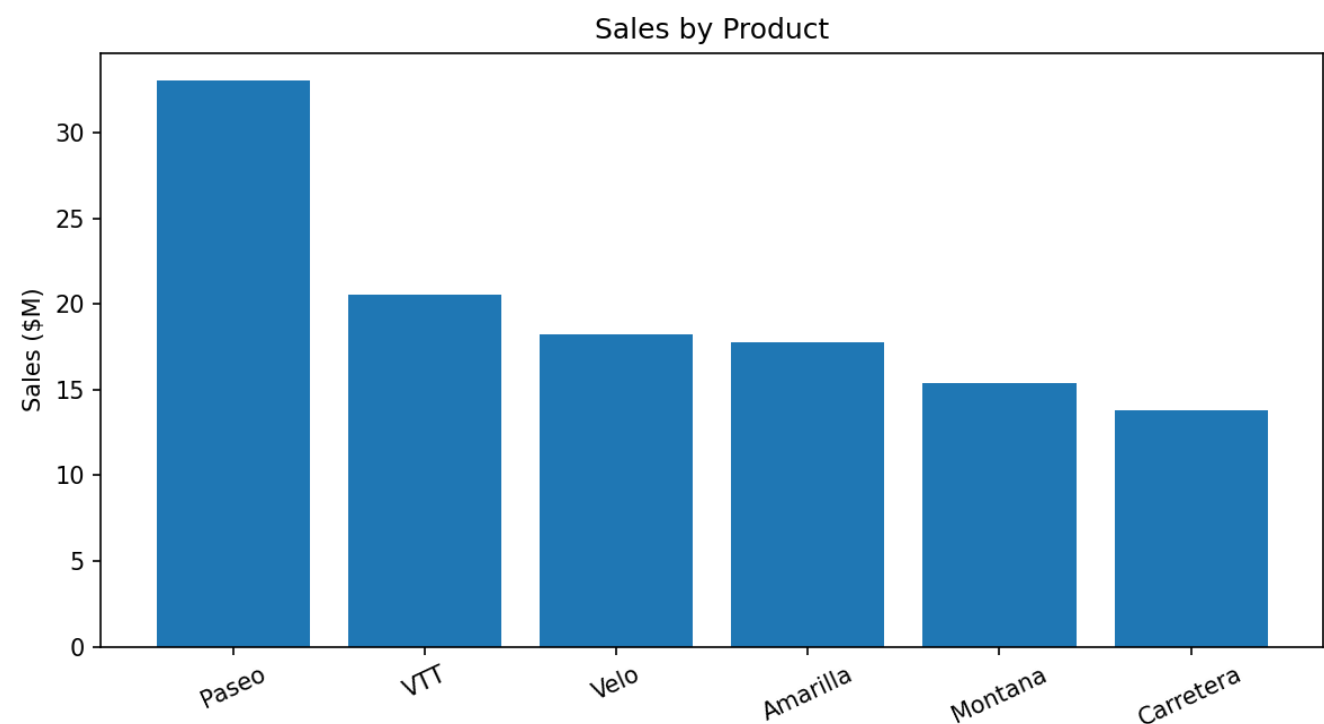
Item	Result
Rows	700
Columns	16
Duplicate rows	0
Original missing Discount Band values	53
Years	2013 and 2014
Countries	5
Products	6
Customer segments	5

Cleaning steps included standardizing column names, parsing the Date column, checking duplicates, and replacing missing Discount Band values with 'Not Specified' so those records remain available for analysis.

2. Key Performance Indicators

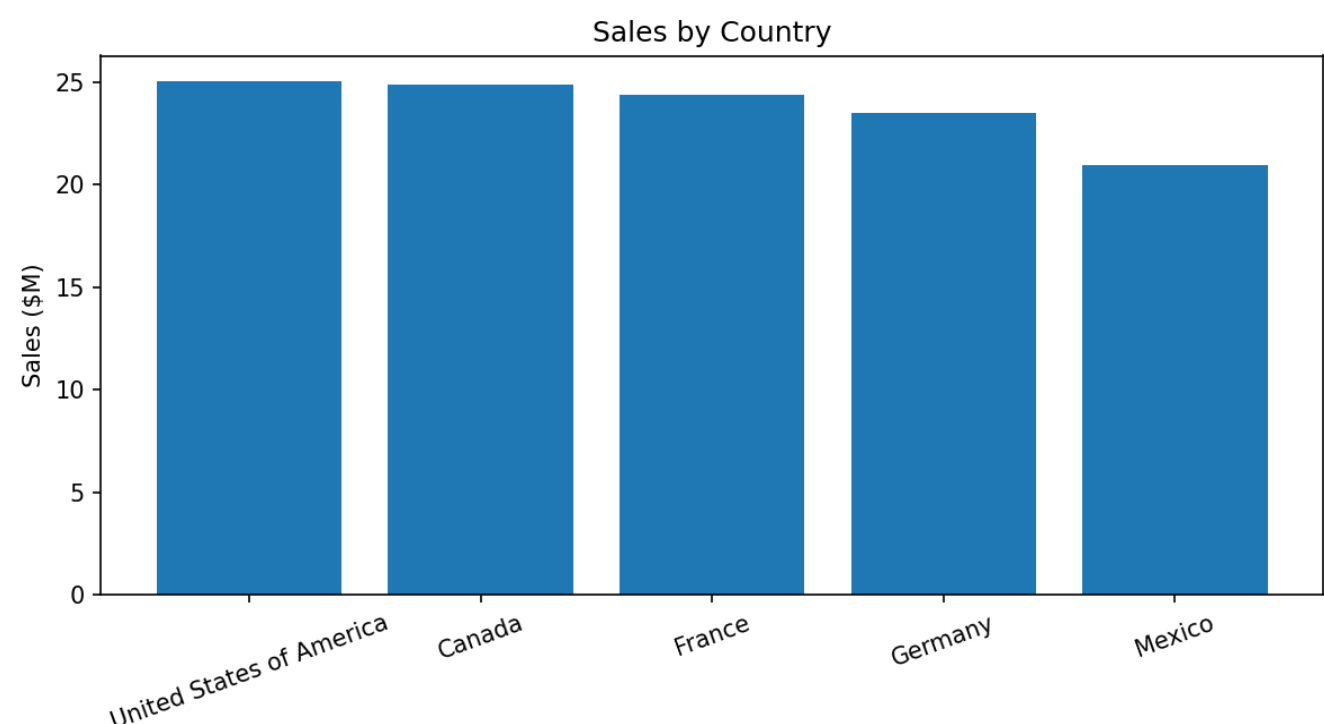
KPI	Value
Total Sales	\$118.73M
Total Profit	\$16.89M
Total Units Sold	1.13M
Overall Profit Margin	14.23%

3. Product Performance



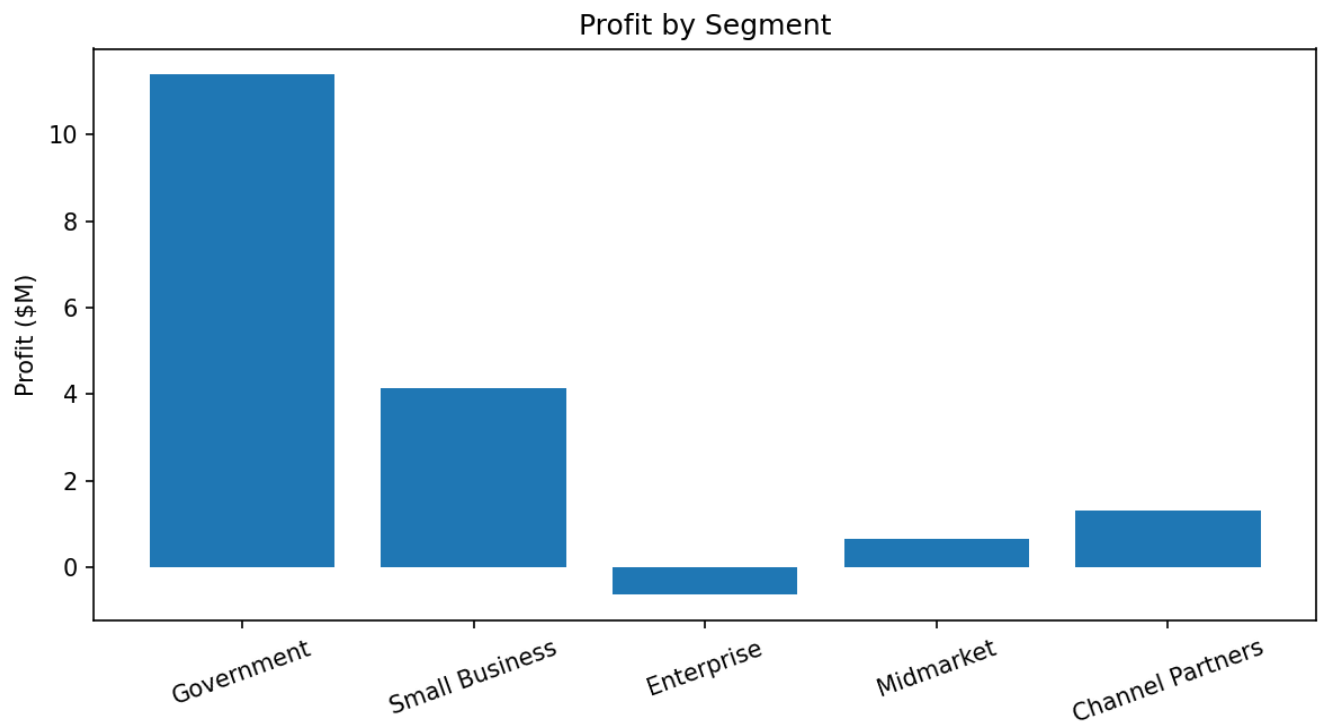
Paseo is the highest-sales product at approximately \$33.01M. Carretera has the lowest sales at approximately \$13.82M. The product comparison helps stakeholders identify the strongest revenue contributors and products that may require further review.

4. Country Performance



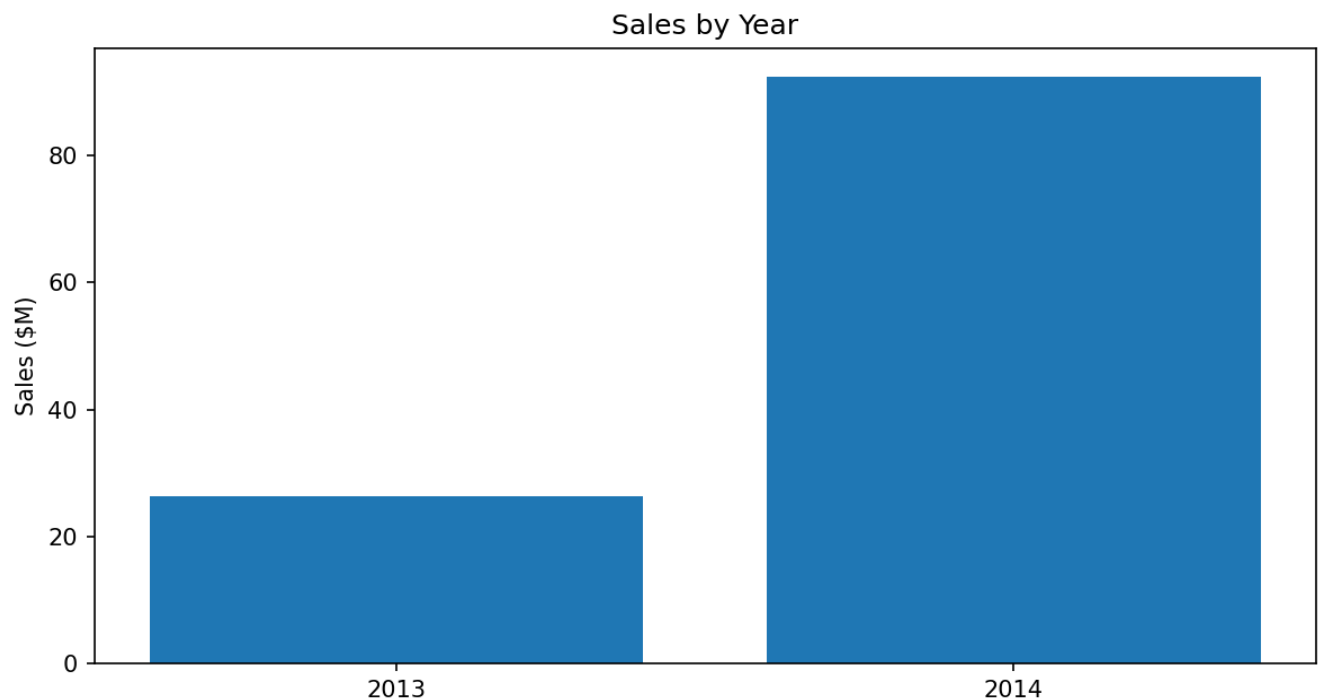
The United States of America has the highest sales at approximately \$25.03M, while Mexico has the lowest sales at approximately \$20.95M. Profitability differs from sales ranking, so stakeholders should consider both revenue and profit rather than sales alone.

5. Segment Performance



Government is the largest segment, generating approximately \$52.50M in sales and \$11.39M in profit. The Enterprise segment records negative total profit of approximately \$0.61M, making it an important area for further investigation.

6. Time Trend



Sales increased strongly from approximately \$26.42M in 2013 to \$92.31M in 2014. The highest-sales month is October, with approximately \$21.67M in sales.

7. Stakeholder Insights & Recommendations

- Prioritize high-performing products such as Paseo while reviewing lower-sales products for pricing, promotion or demand-related issues.
- Use both sales and profit when evaluating countries because the highest sales market is not necessarily the highest-margin market.
- Protect the Government segment because it is the strongest overall contributor to sales and profit.
- Investigate the Enterprise segment to understand why its total profit is negative despite meaningful sales.
- Use the monthly trend to plan inventory, promotions and staffing around stronger sales periods.
- Continue using interactive Power BI filters so stakeholders can drill into country, product, year and segment combinations.

8. Conclusion

The end-to-end analysis shows a business with approximately \$118.73M in sales, \$16.89M in profit and a 14.23% overall profit margin. Paseo is the leading product by sales, Government is the dominant segment, and 2014 contributes substantially more revenue than 2013. At the same time, the negative Enterprise profit highlights an area requiring management attention. The accompanying Excel analysis, Python code, SQL queries and Power BI dashboard provide complementary views of the same dataset and support stakeholder-oriented decision making.